

CIS270A Project 3

This application will require the use of array techniques.

A physical file called GRADEBOOK contains records that track student scores. Each student has 25 scores. The physical file has already been defined as shown below and also contains data. This file is in the library RPGIV_5THS which is on your library list.

```
A* Wexler University Student Exam File
A                                     UNIQUE
A      R EXAMREC
A      STUNO          9S 0          TEXT('Student Number')
A      SFNAME         10          TEXT('First Name')
A      SLNAME         15          TEXT('Last Name')
A*Twenty five scores per student, each is a 3-digit whole number
A      SCORES         75A
A      K STUNO
```

To familiarize yourself with this data, use `RUNQRY QRYFILE ( (GRADEBOOK) )` within the green screen environment and/or download the data to an Excel spreadsheet.

1. Each student's scores need to be averaged. Utilize a runtime array in your solution, along with %XFOOT. The slide "Runtime Arrays and Externally Described Files" is a good reference.
2. Each student should then receive a letter grade. Do NOT code "if" logic for this part of the solution. Utilize compile-time related alternating format arrays for this part of the solution. Use the grading scale shown on the syllabus for this class. Once you have the student's average, use it to look up the corresponding grade in the compile-time data.
3. Design your own output report, including the elements shown below. Be sure to include appropriate editing. As with any program in the course, you will need to utilize some techniques learned in earlier chapters. Use DDS to externally define this report, preferably designed with the RDi Report Designer feature.
 - a. Headings
 - i. An appropriate title
 - ii. Your name
 - iii. The current date, edited
 - iv. The page number
 - v. Column headings for your detail lines
 - b. Detail lines
 - i. Student number
 - ii. Student name as one field with one space between first and last name, such as MARY SMITH
 - iii. The student's average
 - iv. The student's letter grade

Continue to include commenting as outlined in Project 1. Do your housekeeping and send an email when your work is ready to be evaluated on the system. Keep improving your productivity by using the RDi software for coding and compiling!